簡易
Gamut-Mapping処理

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やりたいこと

- RGB立体を使ってカラー画像のY維持型のGamut-Mappingを実現
- 処理はYCrCb空間でのSaturation_UP
- 環境はPython3.8.10＋openCV4.4.0

処理方法

C:¥Users¥kanamori¥VSC_py38¥S_UP1.py

RGB画像

rgb_YCrCb
(colimg)

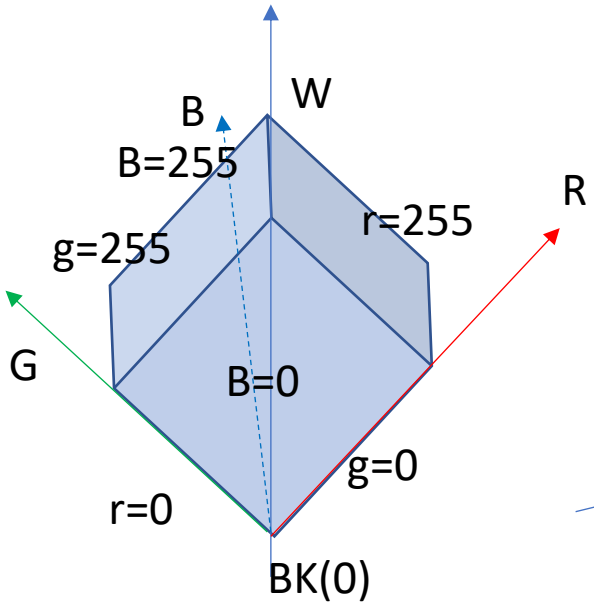
YCrCb_rgb_sup
(colimg,
out_gamut_proc,
sat_up):

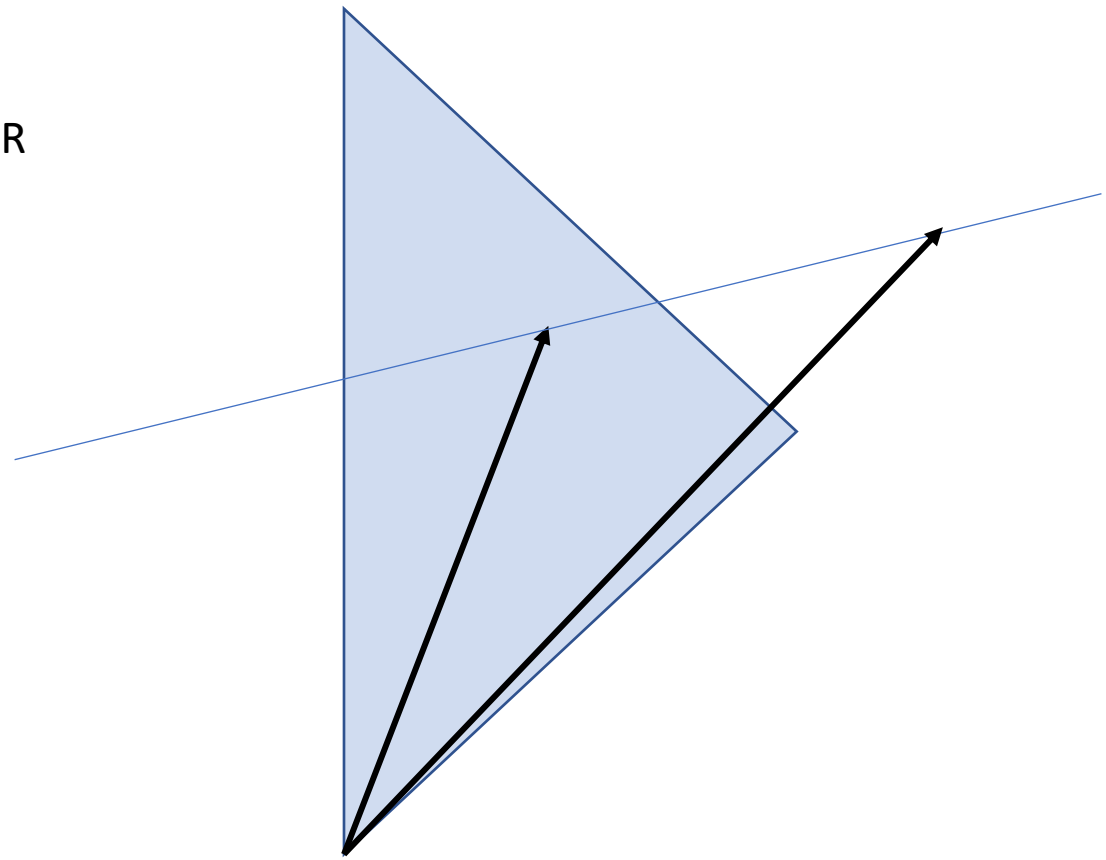
RGB画像

yy = 0.3*r + 0.6*g + 0.1*b
cr = (r - yy)/1.4 + 127
cb = (b - yy)/1.8 + 127

r = (cr - 127) * 1.4 * sat_up + yy
b = (cb - 127) * 1.8 * sat_up + yy
g = (yy - 0.3*r - 0.1*b) / 0.6

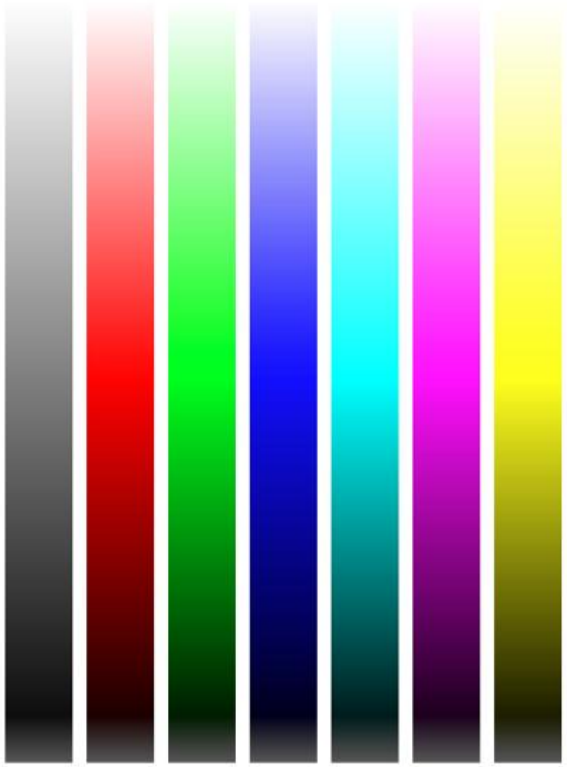
計算方法



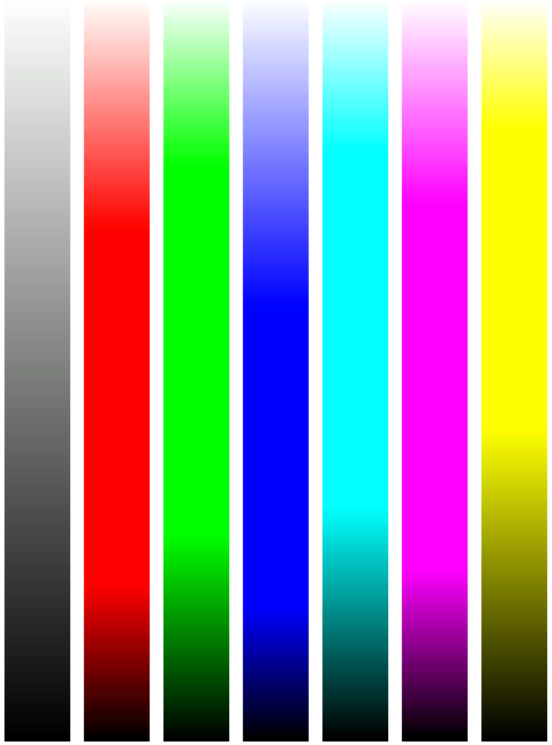


Sx3Mapping結果

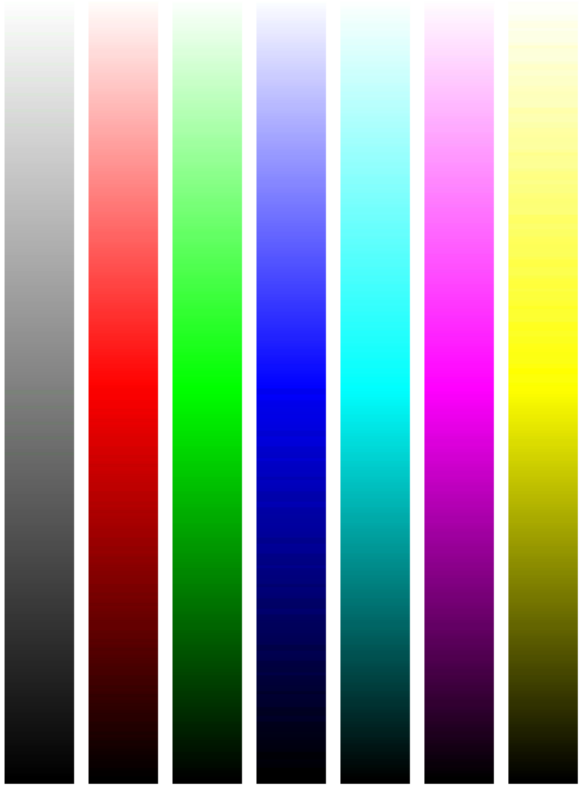
■カラーバーは全てGmaut外に出るが等Y輝度の位置にマッピングされる
■CLIPPINGでは色相変化は無いが階調に乏しい結果



Original Image

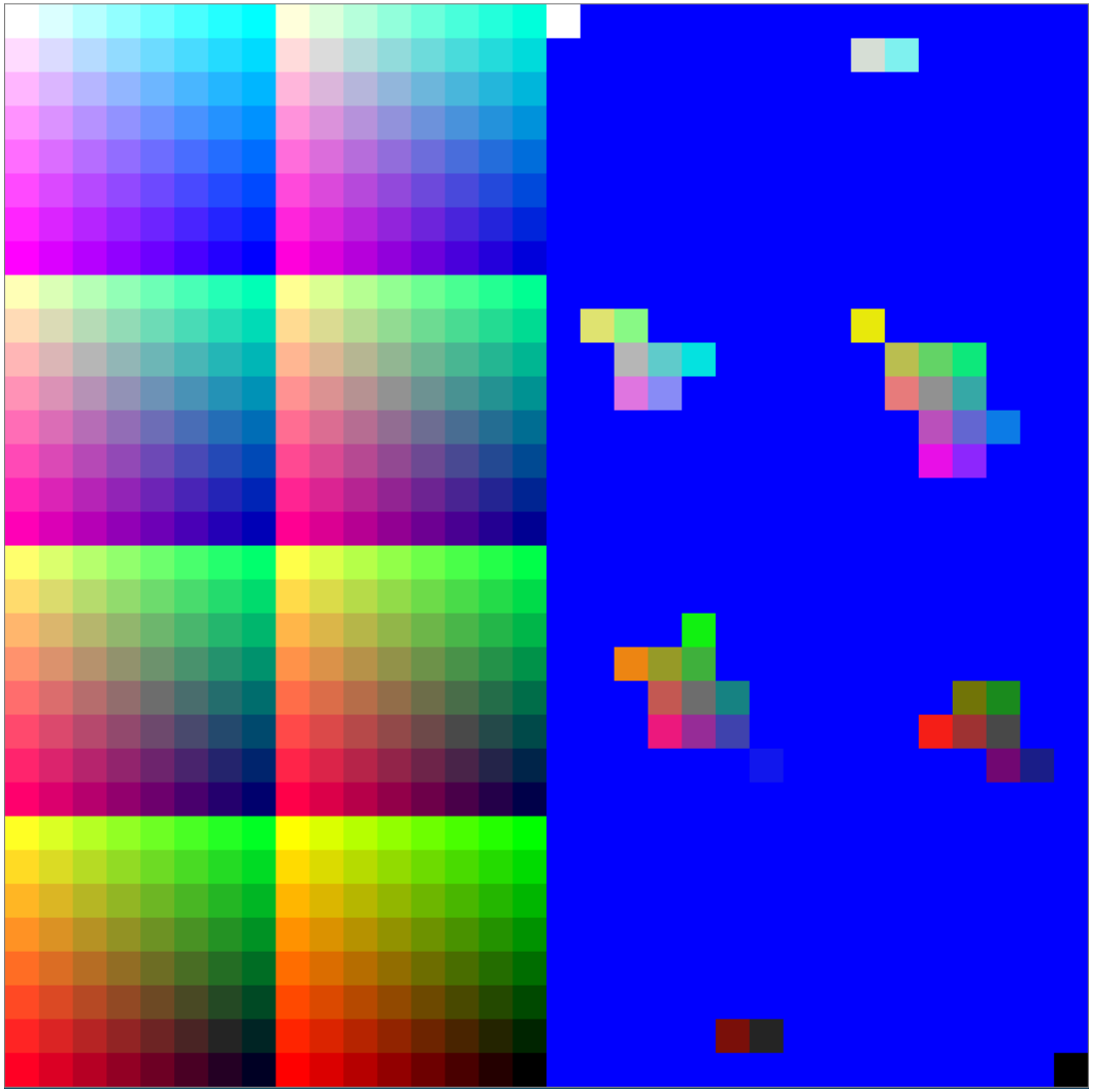


Sx3　CLIPPING



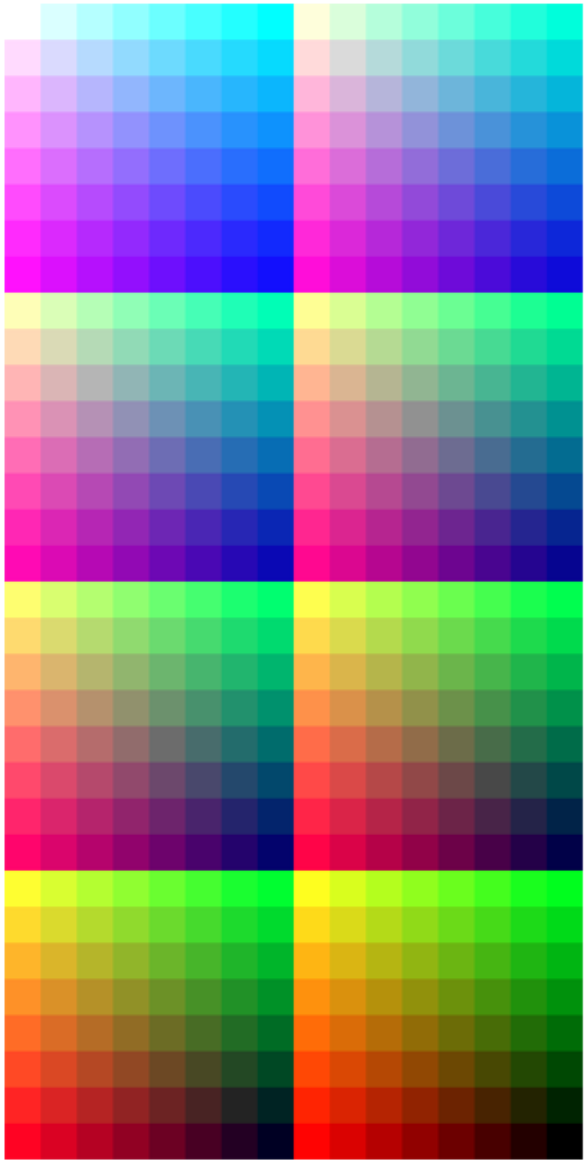
Sx3　GAMUT-MAPPING

Sx3処理結果

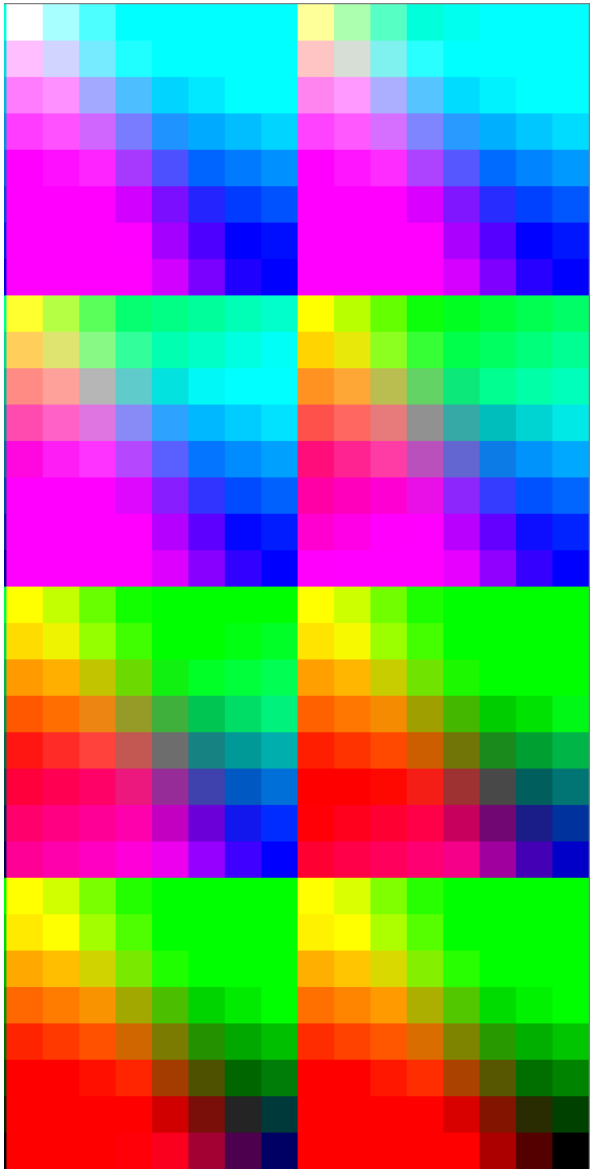


Original Image　　Out of Gamut Area (Blue)

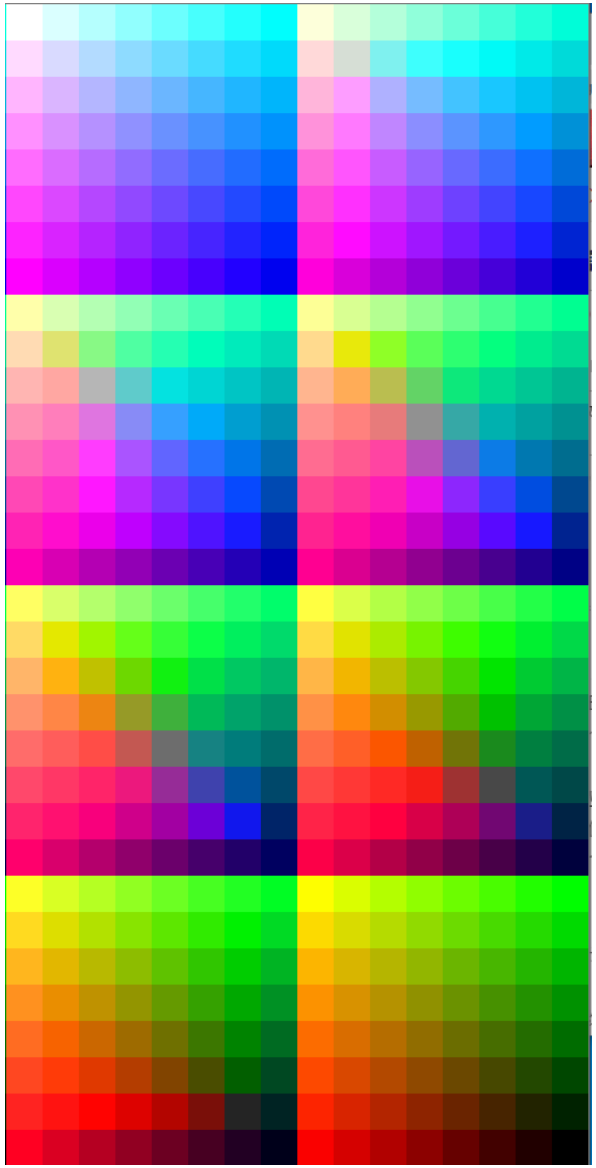
Sx3処理結果



Original Image



Sx3　CLIPPING



Sx3　GAMUT-MAPPING

Sx3処理結果

Original Image

Out of Gamut Area (Blue)

Sx3処理結果

■Tulips画像は原画像がほぼRGB色域の彩度限界まで使っているため彩度UPを実施するとCLIPPINGでは大きく色が変化
■Gamut-Mappingするとほとんど彩度UPの効果が見えない

Original Image

Sx3　CLIPPING

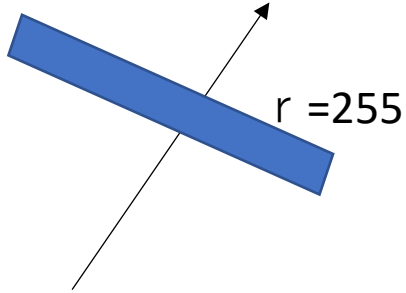
Sx3　GAMUT-MAPPING

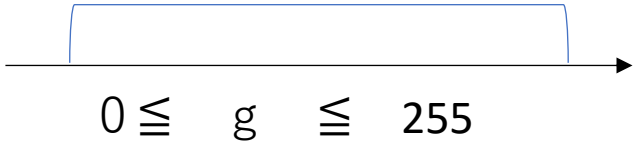
ノイズ改良

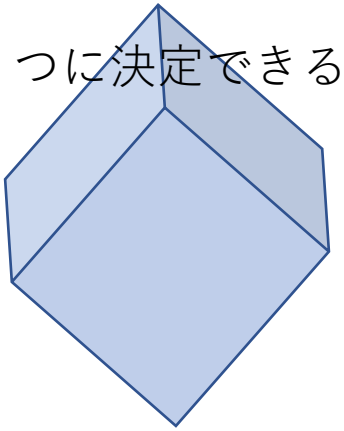
■MappingするRGB表面の平面を決定する際に０≦ｋ≦１の条件だけでは複数の面候補が出てくる

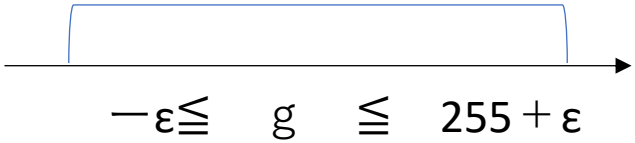
■交点座標のrgbがすべて０≦RGB≦255内に存在する条件で１つに決定できるがその際に誤差が発生して誤った面が選択されてノイズ発生

■少し条件を緩くして解決









Sx3処理結果

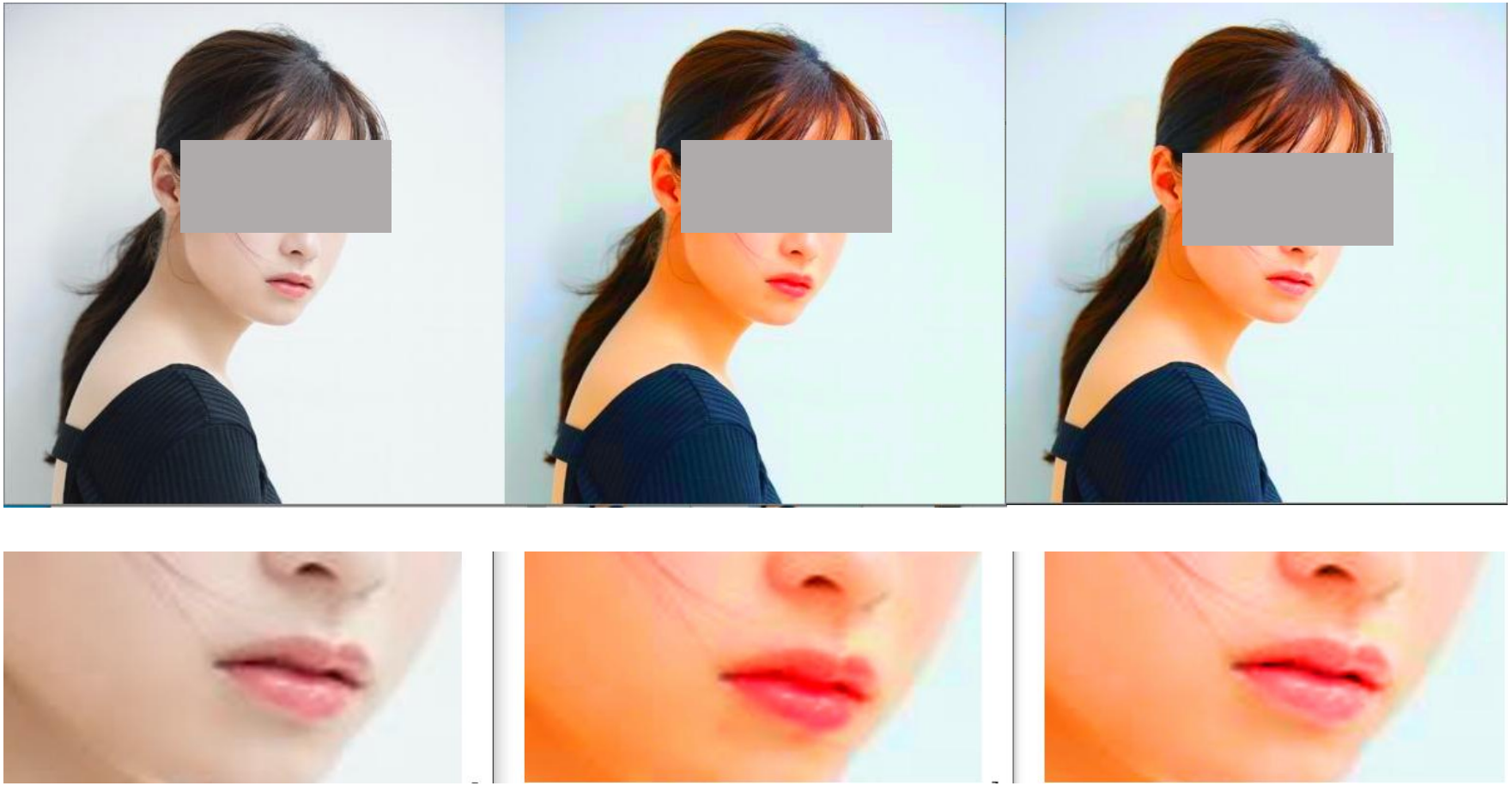
Original Image

Out of Gamut Area (Blue)

Sx3処理結果

Original Image　　Sx3　CLIPPING　　Sx3　GAMUT-MAPPING

Sx3処理結果



Original Image　　Sx3　CLIPPING　　Sx3　GAMUT-MAPPING

Sx3処理結果

Original Image

Sx3　CLIPPING

Sx3　GAMUT-MAPPING